

A 60-year-old woman is scheduled for a right total knee replacement for severe osteoarthritis under general anaesthesia.

A. Discuss the role of regional anaesthesia (RA) techniques for intraoperative and post-operative analgesia in this patient (50%).

Sample Answer

Regional anaesthesia may be useful as **part of a multi-modal analgesia** approach in this patient, who is scheduled for Right TKR under GA.

Intra-op

- **Central neuraxial blocks** may be used to provide complete anaesthesia and analgesia, but at increased risk of intra-op hypotension when administered together with GA.
- **Peripheral nerve blocks** can reduce intra-op opioid usage, thereby reducing opioid-related adverse effects. Benefits include:
 - Reduced PONV incidence
 - Reduced risk of post-op respiratory depression
 - Quicker recovery from GA
 - Earlier resumption of oral intake
- **Potential risks include:** block failure necessitating conversion to GA, inadequate block duration for prolonged surgery, post-dural puncture headache, spinal/ epidural haematoma/ abscess, and nerve injury.

Post-op

Considerations for using RA for post-op analgesia are more complex, due to the need to:

- Facilitate ambulation without motor weakness
- Adequacy of analgesia to allow cooperation with post-op physiotherapy
- Duration of analgesia should be sufficiently long to cover for post-op physiotherapy

Single shot femoral/ obturator/ popliteal nerve block cause motor block, and may have limited utility for fast-track TKR surgery with early patient mobilization. It may also mask and delay the diagnosis/ treatment of surgery-related nerve injury (esp common peroneal nerve).

A true adductor canal block is motor-sparing, but the nerve to vastus medialis should be identified and blocked in order to provide analgesia to the knee joint. Other motor-sparing blocks which may be used in combination with adductor canal block are geniculate nerve blocks and IPACK block.

Alternative techniques which may be useful to augment post-op analgesia include:

- **Peri-neural catheters** into **adductor canal** for continuous LA infusion
- **Infiltrative LA cocktail** technique by the surgeons (with or without catheter in-situ)

At present, there is no conclusive evidence on the single best RA technique for TKR analgesia. It is therefore important to discuss with surgeon and patient on what constitutes acceptable outcomes (balancing benefits with the potential risks of RA) prior to surgery. In addition to RA, multi-modal analgesia should also be used intra-op and post-op for TKR surgery.